

RNE2026 HW4 Report

Object Detection and Semantic Segmentation

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1 Introduction

In this homework, I implemented two YOLO-based perception tasks in the PROS Twin final project environment. The first task is object detection, where the model detects *bear* and *knob* using bounding boxes. The second task is semantic segmentation, where the model segments *road* and *bridge* regions.

The workflow includes data collection in PROS Twin, dataset labeling in Roboflow, YOLO model training, and testing the trained models in Foxglove through the ROS2 YOLO integration package.

2 Roboflow Project Links

- Object Detection Project: [RNE Object Detection Roboflow Project](#)
- Semantic Segmentation Project: [RNE Semantic Segmentation Roboflow Project](#)

3 Dataset Configuration

3.1 Object Detection Dataset

The object detection dataset contains images collected from the final project scene in PROS Twin. The target classes are *bear* and *knob*. Each object is labeled using a bounding box.

Item	Setting
Task type	Object Detection
Classes	bear, knob
Number of images	82
Training set	57 images, 70%
Validation set	16 images, 20%
Test set	9 images, 11%
Annotation format	YOLO bounding box format
Augmentation	No augmentations were applied

Table 1: Object detection dataset configuration.

3.2 Semantic Segmentation Dataset

The semantic segmentation dataset contains images collected from the same PROS Twin environment. The target classes are *road* and *bridge*. Instead of bounding boxes, each target region is labeled using polygon masks.

Item	Setting
Task type	Semantic Segmentation
Classes	road, bridge
Number of images	82
Training set	58 images, 71%
Validation set	16 images, 20%
Test set	8 images, 10%
Annotation format	YOLO segmentation format
Augmentation	No augmentations were applied

Table 2: Semantic segmentation dataset configuration.

4 YOLO Training Configuration

4.1 Object Detection Training

For object detection, I trained a YOLO detection model using the exported Roboflow dataset. The model was trained to detect the bear and knob objects in the final project scene.

Item	Setting
Model type	yolo11n.pt
Task	Object Detection
Epochs	50
Batch size	8
Image size	640
Dataset	RNE object dataset exported from Roboflow
Best weight	detection.pt

Table 3: Object detection YOLO training configuration.

4.2 Semantic Segmentation Training

For semantic segmentation, I trained a YOLO segmentation model using the exported Roboflow segmentation dataset. The model was trained to predict the pixel-level regions of road and bridge.

Item	Setting
Model type	yolo11n-seg.pt
Task	Semantic Segmentation
Epochs	50
Batch size	4
Image size	640
Dataset	RNE segmentation dataset exported from Roboflow
Best weight	segmentation.pt

Table 4: Semantic segmentation YOLO training configuration.

5 Dataset Diversity

To improve the generalization ability of the models, I collected images from different view-points and distances in the final project scene. The dataset includes objects and regions under different positions, scales, and viewing angles.

For object detection, the bear and knob objects were captured from multiple camera angles. Some images contain clear frontal views, while others contain side views or objects located near the edge of the camera frame. This helps the model learn the appearance of the objects under different conditions.

For semantic segmentation, the road and bridge areas were labeled in scenes with different camera positions and route layouts. Since road and bridge regions may appear with different shapes depending on the viewpoint, using diverse images helps the model produce more stable segmentation masks during testing.



Figure 1: Example object detection annotation for bear and knob in Roboflow.

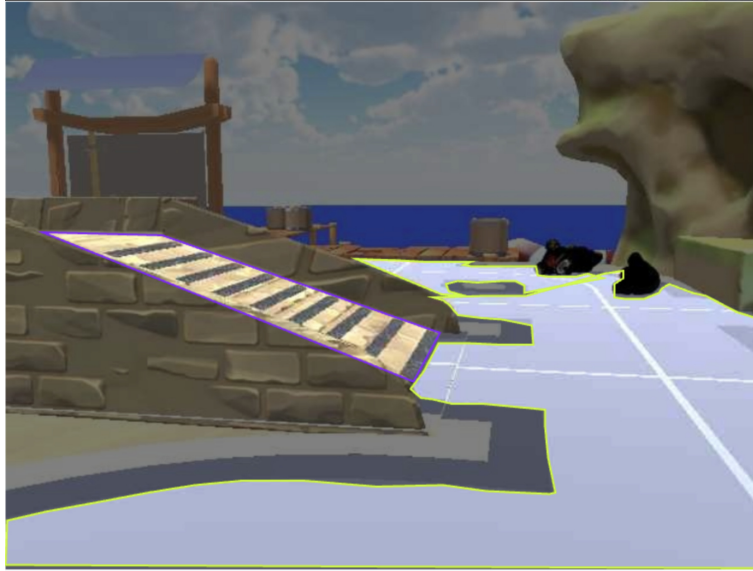


Figure 2: Example semantic segmentation annotation for road and bridge in RoboFlow.

6 Navigation Strategy in the Final Project

During data collection and testing, I controlled the vehicle manually in the final project scene. The main strategy was to move the vehicle slowly and keep the camera facing the target area as much as possible. This helped collect clearer images and also made the testing video easier to evaluate.

For object detection, I navigated the vehicle toward areas where the bear and knob were likely to appear. When the target object appeared in the camera view, I adjusted the vehicle position and camera direction to keep the object visible for several seconds. This allowed the YOLO detection model to continuously display bounding boxes in Foxglove.

For semantic segmentation, I focused on routes where the road and bridge were clearly visible. I avoided moving too fast because fast motion could make the camera view unstable. By keeping the vehicle movement smooth, the segmentation result became easier to observe in the recorded Foxglove video.

The overall navigation strategy can be summarized as follows:

1. Start from the final project scene and switch to the proper control mode.
2. Move the vehicle slowly to search for target objects or target regions.
3. Adjust the camera direction to keep the target visible.
4. Record the YOLO output in Foxglove for around 30 to 60 seconds.
5. Repeat the process for both object detection and semantic segmentation models.

7 Testing Result

After training, I placed the trained model weights into the ROS2 YOLO integration package and modified the model path in the YOLO node. The detection and segmentation results were visualized in Foxglove.

The object detection model was able to detect the target objects using bounding boxes. The semantic segmentation model was able to predict mask regions for road and bridge. The testing results were recorded as two demo videos: `detection.mp4` and `segmentation.mp4`.

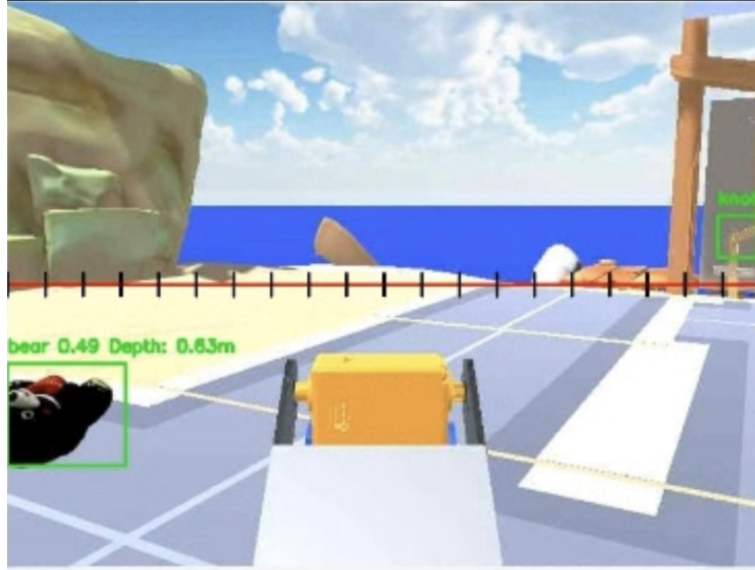


Figure 3: Object detection testing result in Foxglove.

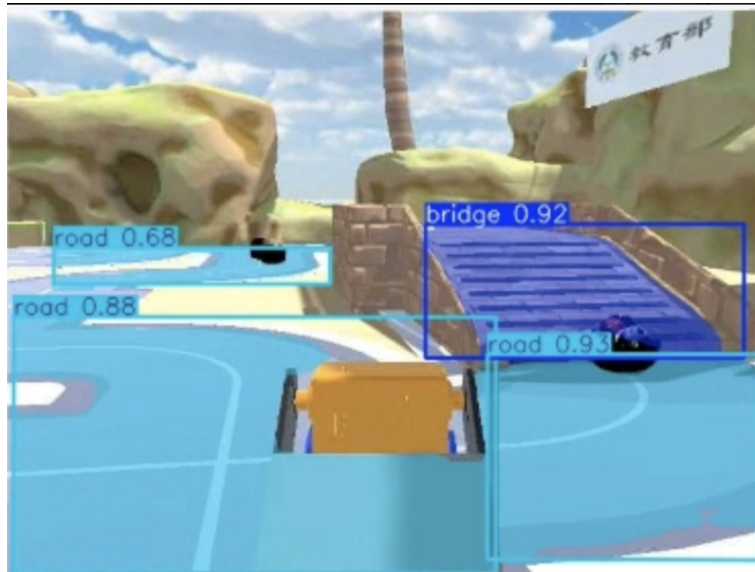


Figure 4: Segmentation result in Foxglove.

8 Conclusion

In this homework, I completed the full workflow of collecting data, labeling images, training YOLO models, and testing the models in the PROS Twin environment. Through this process, I learned the difference between object detection and semantic segmentation. Object detection predicts bounding boxes for target objects, while semantic segmentation predicts the target areas at the pixel level.

The final trained models and demo videos show that YOLO can be integrated with the robot simulation environment to perform real-time perception tasks. The quality of the

result is strongly affected by dataset diversity, annotation quality, and the stability of the camera view during testing.